

READING PASSAGE 3

The Physics of Earthquakes

A The Physics of Earthquakes has emerged as one of the most significant topics in contemporary academic discourse. The complexity and multifaceted nature of this subject have attracted the attention of researchers, policymakers, and practitioners across multiple disciplines, all seeking to understand its implications for the future.

B Historically, scholarly understanding of this topic was limited by both theoretical frameworks and available methodologies. Early researchers operated within relatively narrow conceptual boundaries, and their findings, while groundbreaking for their time, often failed to capture the full complexity of the phenomena they sought to explain. The development of new research tools and analytical techniques in the latter half of the twentieth century fundamentally transformed the landscape, enabling scholars to investigate questions that had previously been considered beyond the reach of empirical inquiry.

C Contemporary research has identified several critical factors that determine outcomes in this domain. The interplay between environmental conditions, technological capabilities, and social organisation creates a complex adaptive system in which small changes can produce disproportionately large effects. Understanding these dynamics requires analytical approaches that move beyond simple cause-and-effect reasoning to embrace the principles of complexity science and systems thinking.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E The practical application of research findings has proven more challenging than many initially anticipated. The translation of academic knowledge into actionable policy recommendations requires navigating a complex landscape of political constraints, economic interests, and social dynamics that academic models often simplify or overlook. Implementation failures in several high-profile initiatives have underscored the gap between theoretical understanding and practical capacity, prompting calls for more engaged and context-sensitive approaches to research.

F Looking ahead, the field stands at a critical juncture. Emerging technologies offer unprecedented opportunities for data collection and analysis, potentially enabling researchers to address questions that have long remained beyond the reach of empirical investigation. At the same time, the accelerating pace of social and technological change means that the phenomena under study are themselves evolving, requiring theoretical frameworks that can accommodate dynamism and discontinuity rather than assuming stability and equilibrium.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 1 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 2

Choose the correct letter, A, B, C or D.

- 2 According to the text, what role do structural factors play?
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone

Question 3

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 3 Early researchers had access to the same technologies as modern scholars.

Questions 4–5

Choose the correct letter, A, B, C or D.

- 4 What does the passage identify as a significant challenge to current understanding?
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications
- 5 What does the passage identify as a significant challenge to current understanding?
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications

Question 6

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 6 The passage suggests that structural factors are less important than individual choices.

Questions 7–8

Choose the correct letter, A, B, C or D.

- 7 According to the passage, what is one of the key factors identified by researchers?
- A. minimal impact
 - B. creating underlying conditions for action
 - C. determining individual choices directly
 - D. irrelevant to outcomes
- 8 Which of the following best describes the author's view on future developments?
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 9

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 9 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Questions 10–13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 10 The author notes that researchers have called for more _____ approaches to investigation.
- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.
- 12 The landmark study analysed data spanning a period of _____ years.
- 13 Case studies have documented the effects of _____ change on the system.